

Hyperparameter Training

QMML Society



Let's imagine baking a cake

Imagine you have a cake of box mix.

What "settings" on the box do you control to get a perfect cake?



Making a ML model is like baking a cake



Recipe

This is your **Model** (e.g., a Neural Network recipe).

Ingredients

This is your **Data** (e.g., flour, sugar, eggs).

Knobs

These are your **Hyperparameters** (Oven Temp, Bake Time).

Hyperparameters in Baking a Cake

Bake Time

Recipe 1 (original)	35 minutes
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Recipe 2	45 minutes
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Hyperparameters in Baking a Cake

Bake Time

Recipe 1 (original)	35 minutes
------------------------	------------

Recipe 2	45 minutes
----------	------------

Recipe 3	40 minutes
----------	------------

Hyperparameters in Baking a Cake

Bake Time

**Bake
Temperature**

**Recipe 1
(original)**

35 minutes

350

Recipe 2

45 minutes

Recipe 3

40 minutes

Hyperparameters in Baking a Cake

Bake Time

**Bake
Temperature**

**Recipe 1
(original)**

35 minutes

350

Recipe 2

45 minutes

325

Recipe 3

40 minutes

Hyperparameters in Baking a Cake

Bake Time

**Bake
Temperature**

**Recipe 1
(original)**

35 minutes

350

Recipe 2

45 minutes

325

Recipe 3

40 minutes

330

Hyperparameters in Baking a Cake

	Bake Time	Bake Temperature	Number of eggs
Recipe 1 (original)	35 minutes	350	2
Recipe 2	45 minutes	325	
Recipe 3	40 minutes	330	

Hyperparameters in Baking a Cake

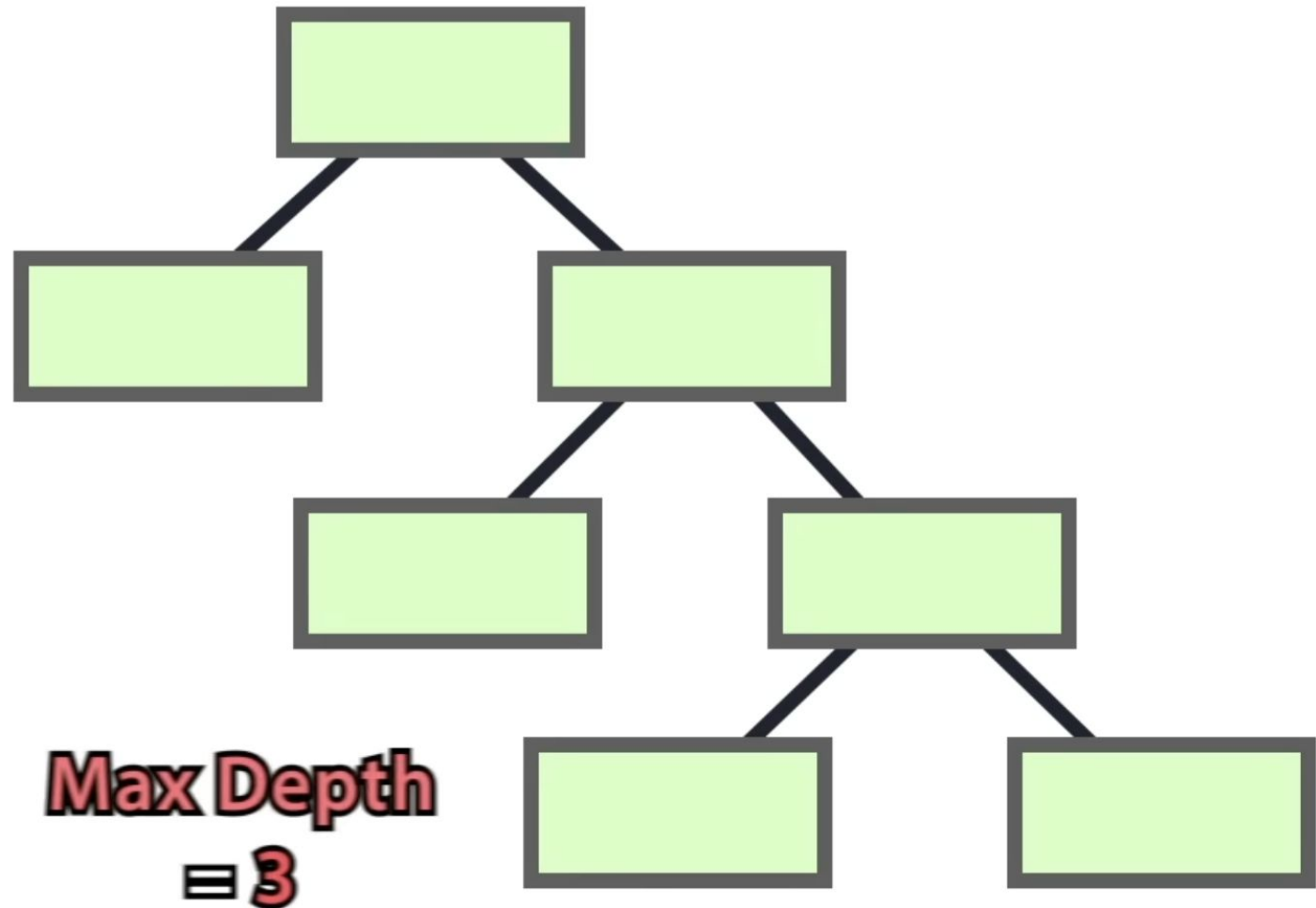
	Bake Time	Bake Temperature	Number of eggs
Recipe 1 (original)	35 minutes	350	2
Recipe 2	45 minutes	325	3
Recipe 3	40 minutes	330	3

Hyperparameters in Baking a Cake

	Bake Time	Bake Temperature	Number of eggs	Mayo
Recipe 1 (original)	35 minutes	350	2	Yes
Recipe 2	45 minutes	325	3	No
Recipe 3	40 minutes	330	3	Yes

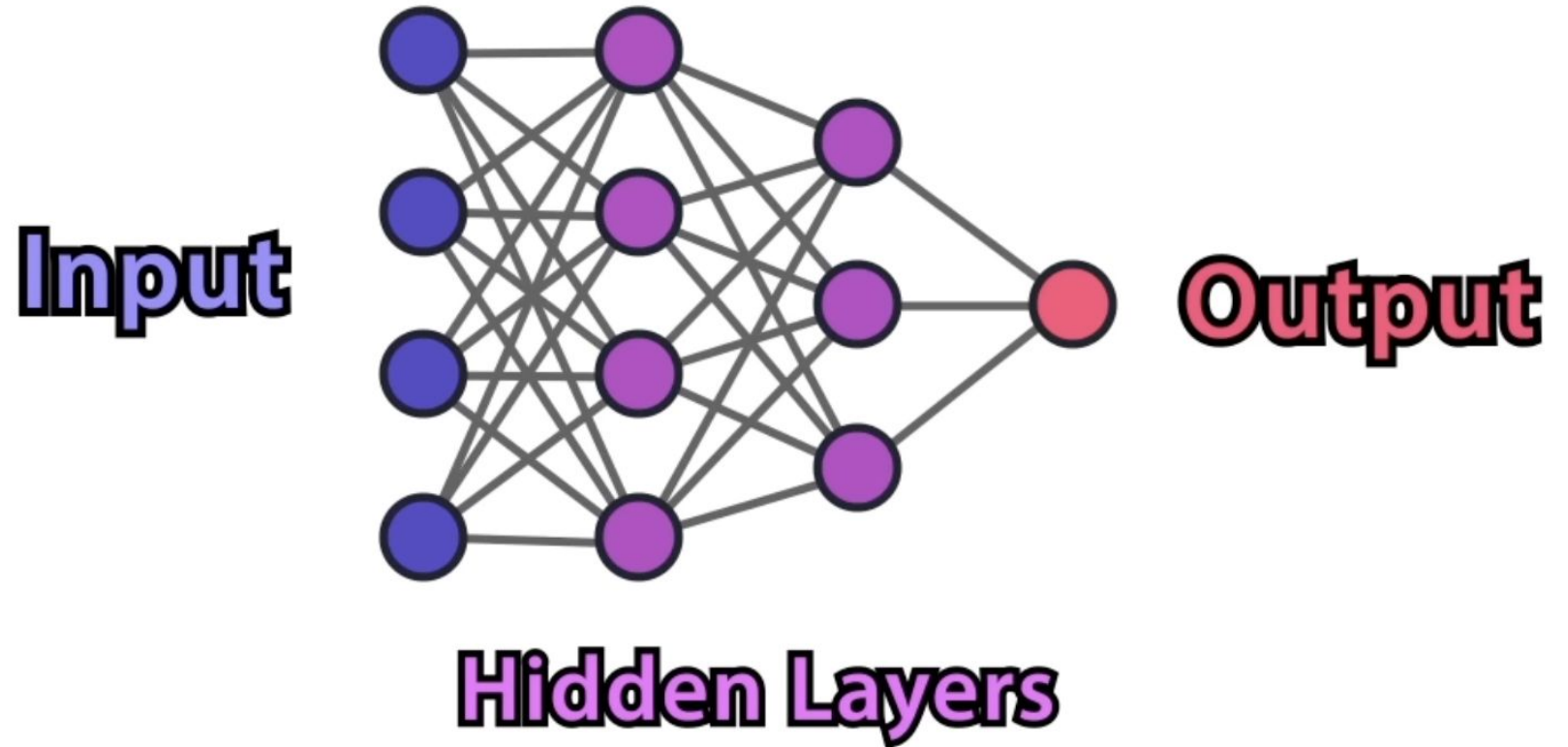
Common Hyperparameters in ML

Maximum depth
in a decision
tree



Common Hyperparameters in ML

Number of
Hidden Layers in
a Neural
Network



Common Hyperparameters in ML

Learning Rate in Linear
Regression



Common Hyperparameters in ML

Batch Size



What are Hyperparameters?

They are the 'settings' you choose **before** you start training (or baking).

Too hot? Your cake burns. This is called **Overfitting**.

Too cool? Not fully cooked. This is called **Underfitting**.

The GOAL: Find the 'just right' settings for the perfect model.

What's the difference?

Hyperparameters (The Knobs)

YOU set these.

They are the settings outside the model that control how it learns.

e.g., Oven Temperature, Bake Time, Learning Rate.

Parameters (The Cake)

The Model learns these.

They are the settings inside the model that are the result of learning.

also known as knowledge of the model.

Choosing the right parameters is very important!!

Can significantly affect model of the performance.

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How to Find the Perfect Recipe

Part 1: Guessing

Method 1: The "Perfectionist" (Grid Search)

You make a list of every single combination you can think of.

- ★ **Temp:** 180°, 190°, 200°
- ★ **Time:** 30m, 35m, 40m
- ★ then you bake 9 cakes ($3 \times 3 = 9$)

Pro: You'll find the best one *in your list*.

Con: Really SLOW.



Method 1: (Grid Search)



	Bake Time	Bake Temperature	Number of eggs	Mayo
Recipe 1 (original)	35 minutes	350	2	Yes
Recipe 2	45 minutes	325	3	No
Recipe 3	40 minutes	330	3	Yes

Method 1: (Grid Search)

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Method 1: (Grid Search) Too Slow!

11 knobs, 54 possibilities

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Method 1: (Grid Search) Too Slow!

Hyperparameter 1	1	3	5	7	9	11
Hyperparameter 2	10	15	20	25	30	35
Hyperparameter 3	0.1	0.4	0.7	1	1.3	1.6
Hyperparameter 4	5	6	7	8	9	10
Hyperparameter 5	2	4	6	8	10	12

You will end up training 6^5 models $\equiv$ **7776**

Method 2: The "Lucky Experimenter" (Random Search)



Instead of trying everything, you just bake (say) 9 cakes with **random settings**.

- ★ e.g., 340° for 37m
- ★ e.g., 350° for 32m
- ★ ...and so on!

Pro: Much faster! Surprisingly effective.

Con: You *might* miss the perfect combo, but you'll get close.

Method 2: The "Lucky Experimenter" (Random Search)

Hyperparameter 1

1 - 9

Hyperparameter 2

10 - 30

Hyperparameter 3

0.1 - 1.3

Hyperparameter 4

5 - 9

Range

I want to try **10 models**
with these specific
ranges.

This method randomly
samples hyperparameter
from range.

Random Search Usually WINS

Why? Because some knobs are far more important than others.

Grid Search wastes time checking unimportant knobs over and over.

Random Search explores more different values for each knob, and is more likely to find a "good enough" setting on the important ones

Limitation of Grid Search and Random Search



[blog.DailyDoseofDS.com](https://blog.dailydoseofds.com)

	Grid Search	Random Search	Bayesian Optimization		
Total Iterations	720	360	100	7x less iterations	✓
Total Run-time	242s	113s	48s	5x less run-time	✓
Best Trial Index	667	221	83	Optimal config. found earlier	✓
Best F1 score	0.93	0.93	0.93	Same score	✓

How to Find the Perfect Recipe

Part 2: The "Smart" Methods

Method 3: The "Smart Guesser" (Bayesian Optimization)

Bayesian approaches, in contrast to random or grid search, keep track of **past evaluation results** which they use to form a probabilistic model.

$$P(\textit{score} \mid \textit{hyperparameters})$$

Method 3: The "Smart Guesser" (Bayesian Optimization)

Step 1: Bake

Bake one random cake. (e.g., 340° for 30m).

Step 2: Taste

You taste it. "Score: 6/10. A bit dry and not sweet."

Step 3: Guess Smart

Your brain "models" this. "OK, next I'll try **less** time and **more** sugar."

Bayesian Optimization -
pick a **prior**
belief and then search
the parameter space by
enforcing & updating that
prior belief during training

Early Stopping: "Peeking in the Oven"

Why bake 20 full cakes if you can tell they're burning after 10 minutes?

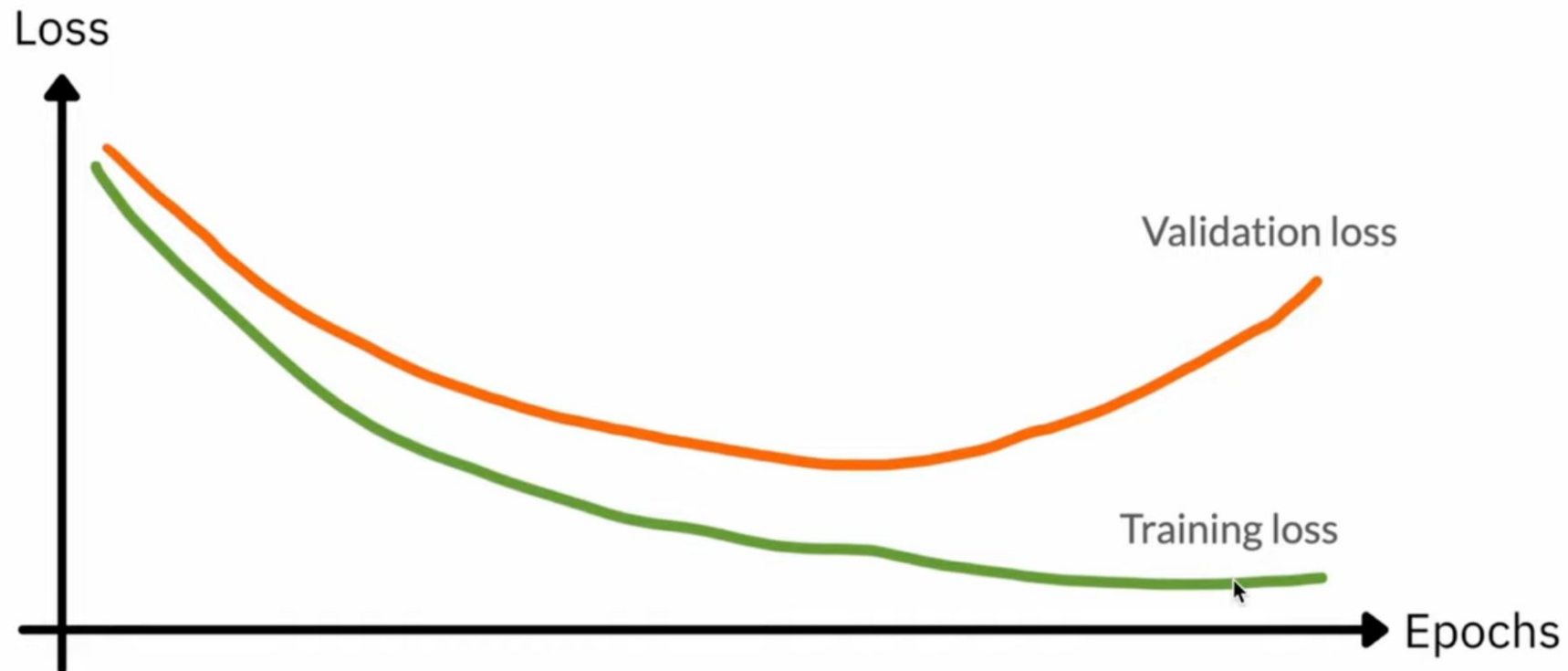
Early Stopping is a simple, powerful trick:

1. Check the "score" every few minutes (epochs).
2. If the score isn't improving, just STOP training.
3. Throw that cake away and try a new recipe!



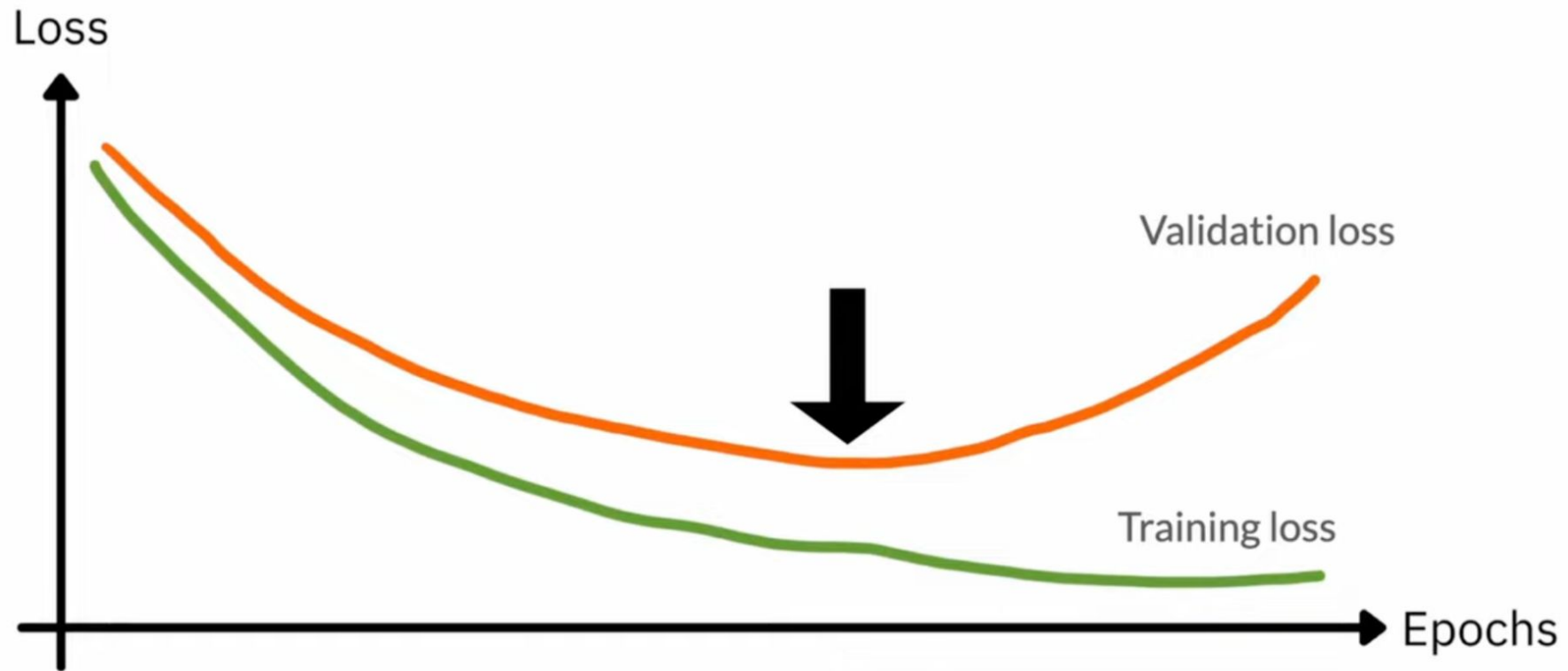
Early Stopping

Stop the learning once the validation error is minimum.



Early Stopping

Stop the learning once the validation error is minimum.



Tips for Tuning your model



Start "Coarse" then "Fine": Do a **wide** random search first. Then, "zoom in" with a **narrow** search (Grid or Bayesian) on the best area.



Use a Log Scale! This is a big secret. For knobs like ``learning_rate``, search `[0.1, 0.01, 0.001]`, NOT `[0.1, 0.5, 0.9]`.



Start with Random Search: Don't overcomplicate it. Random Search + Early Stopping is simple and works 90% of the time!

What to Remember (TL;DR)



Hyperparameters are the 'settings' you set **before** training.



Tuning them is like finding the perfect **cake recipe**.



Random Search is usually better and faster than Grid Search.



Bayesian Optimization is a "smart" guesser that learns from its tests.